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$x_i x_j$
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?
C_1
C_2
w

$$y = \mathbf{w}^T x \tag{1}$$

S_W
S_B

$$\mathbf{S}_W = \sum_{n \in C_1} (\mathbf{x}_n - \mathbf{m}_1)(\mathbf{x}_n - \mathbf{m}_1)^T + \sum_{n \in C_2} (\mathbf{x}_n - \mathbf{m}_2)(\mathbf{x}_n - \mathbf{m}_2)^T \tag{2}$$

$$J(\mathbf{w}) = \frac{\mathbf{w^T S_B w}}{\mathbf{w^T S_W w}} \tag{3}$$

?

$$\mathbf{w}^* \propto \mathbf{S_W^{-1}(m_1 - m_2)} \tag{4}$$

$$\mathbf{m} = \frac{1}{N} \sum_{k=1}^{K} N_k \mathbf{m}_k, \mathbf{S}_W = \sum_{k=1}^{K} \mathbf{S}_k, \mathbf{S}_B = \sum_{k=1}^{K} N_k (\mathbf{m}_k - \mathbf{m})(\mathbf{m}_k - \mathbf{m})^T \tag{5}$$

?

$$J(\mathbf{W}) = \frac{|\mathbf{W}^T \mathbf{S}_B \mathbf{W}|}{|\mathbf{W}^T \mathbf{S}_W \mathbf{W}|} \tag{6}$$

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$$\mathbf{S_B w_i} = \lambda_\mathbf{i} \mathbf{S_W w_i} \tag{7}$$

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$$\delta_k(x) = -\frac{1}{2} \ln |\Sigma_k| - \frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) + \ln \pi_k \tag{8}$$

$p >$
n
$\mathbf{S_W}$
$\mathbf{S_W}$
$\mathbf{S_W^{-1}}$
?
?
$\mathbf{W}$

$$\mathbf{W^T S_W W = I} \tag{9}$$

$$\mathbf{S}_W = \mathbf{U} \Lambda \mathbf{U}^T, \tag{10}$$

$\mathbf{U}$
Λ
$\mathbf{W} =$
$\mathbf{U}\Lambda^{-1/2}$
$\mathbf{W}$
$\mathbf{S_W}$
$\mathbf{S_W}$
$\mathbf{X}$
$\mathbf{X} =$
$U \Sigma V^T$
Σ
σ_i
τ